

Victory Free WiFi:
Barangay Internet
Project Initiatives



Zamboanga del Sur
Provincial Government

Communication & Information
Services Office (CISO)



Presented to:

HON. DIVINA GRACE C. YU
Provincial Governor

Presented by:

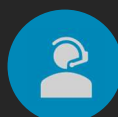
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REPORT

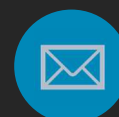
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Abstract

Over the past 4 years, the **Victory Free WiFi** project has transformed digital connectivity across 27 municipalities and more than 400 barangays, bridging the digital divide and empowering communities through accessible internet services.

Project Overview

Launched on April 2021 with the mission to democratize internet access, **Victory Free WiFi** aimed to provide reliable, high-speed wireless connectivity to underserved and geographically isolated areas. By strategically deploying public WiFi hotspots in government centers, schools, health facilities, and public spaces, the project has become a cornerstone of digital inclusion.

Key Achievements

- **Coverage Expansion:** Successfully installed and maintained WiFi infrastructure across 27 municipalities and almost 400 barangays.
- **Community Impact:**
 - Enabled remote learning for thousands of students.
 - Facilitated access to government services and telehealth.
 - Supported local businesses with digital tools and online presence
- **Sustainability:** Leveraged partnerships with BLGUs, NGOs, and private sector stakeholders to ensure long-term viability and maintenance.

Technology & Infrastructure

- Over 600 devices installed across more than 600 sites.
- Deployed solar-powered base stations in off-grid locations.
- Allocates 4Gbps dedicated lease line bandwidth from PLDT-iGate.
- Utilized AirFiber backhaul and Rocket Radio uplinks to reach remote areas.
- Implemented centralized network monitoring and bandwidth optimization.

Outcomes & Future Directions

- **Educational Aide:** Students can easily access the learning materials they need.
- **Digital Literacy Growth:** Increased tech adoption and digital skills among residents.
- **Economic Uplift:** Boosted local economies through e-commerce and online employment.
- **Next Phase:** Plans to expand to additional barangays and integrate smart city applications.

The **Victory Free WiFi** project stands as a model for inclusive digital transformation, proving that with vision, collaboration, and innovation, connectivity can become a public good that uplifts entire communities.

Working Group

1. Management

- **CISO Chief:** Provides logistics support and serves as the overall project manager.
- **Programmer:** Ensures the integrity of installations and compliance with standards.
- **Network Administrator:** Manages network and generates reports to appropriate teams.
- **Master Electrician:** Oversees all electrical installations, ensuring to meet the standards.

2. Deployment

- **Lead Technician:** Conducts site surveys, installations and serves as the team leader.
- **IT Support:** Handles network device cabling and termination.
- **Electrician:** Performs electrical installation, termination, and cabling.
- **Riggers:** Responsible for mounting and installing devices and antennas.
- **Logistic Driver:** Transporting equipment, supplies and personnel to project sites.

3. Monitoring

Monitoring plays a critical role in maintaining the stability, reliability, and performance of the network infrastructure. Their responsibilities include routine inspections, troubleshooting, and coordination with field personnel to ensure uninterrupted public Wi-Fi service.

- **Maintenance Team**
 - Responds to system alerts and coordinates repairs.
 - Oversees the overall health of the network infrastructure.
 - Conducts scheduled maintenance and performance checks.
- **SSFO Facilitators**
 - Serve as on-site correspondents in barangays.
 - Report technical issues and assist with basic troubleshooting.
 - Facilitate communication between local users and the central maintenance team.

Project Sites

The project sites are primarily located within the province of Zamboanga del Sur, spanning from the Data Center to Central Base Stations and Barangay Access Points.

- **1 Data Center** situated at the Provincial Capitol Building
- **4 Central Base Stations** located at:
 - Capitol Building – Pagadian
 - Mt. Palpalan – Pagadian
 - Mt. Dipalutao – Tukuran
 - Kapamanok – San Pablo
 - 70 Municipal Base Stations strategically placed across all municipalities
 - Approximately 600+ Barangay Access Points

References:

- See attached figures and tables
- Please visit our website: www.victoryfreewifi.net
- Project Sites document for detailed geographical locations



Introduction

A. Background

During the COVID-19 pandemic, the Provincial Government of Zamboanga del Sur remained steadfast in its commitment to deliver essential services to its constituents. Among these services, internet access emerged as a critical need—yet many remote communities continued to lack reliable connectivity.

The national government, through the Department of Information and Communications Technology (DICT), initiated Public Wi-Fi Projects. However, only a little over 100 sites were established out of more than 600 barangays across the province.

In response, the Provincial Government of Zamboanga del Sur launched a collaborative effort with technical agencies and other stakeholders to address these connectivity gaps. This initiative, known as the **Province-Wide Internet Project**, aims to provide internet access to every barangay, in alignment with Republic Act 10929, otherwise known as the *Free Internet Access in Public Places Act*, signed by President Rodrigo Roa Duterte.

B. Purpose and Scope

In the context of the COVID-19 pandemic, internet access has become a basic necessity to support government programs, particularly in the education sector. The Department of Education (DepEd) is set to implement distance learning and online modules to promote social distancing and ensure educational continuity.

The project, named **Victory Free Wi-Fi**, is designed to meet this urgent need. It has been implemented province-wide, covering nearly 400 barangays across 27 municipalities, and serving approximately 30,000 users.

The general public can access the network and connect to other local networks or the broader internet through this system.

C. Technical Overview

The network infrastructure primarily utilizes wireless technology operating on the 2.4GHz and 5GHz bands, extending from bridges and base stations to barangay access points.

The network's data center, management, and control systems are housed at the Provincial Capitol Building. Detailed network topologies are provided in the attached documentation.

Users can wirelessly access Wi-Fi hotspot access points within a 200-meter radius, with managed speeds of up to approximately 5 Mbps. Access is regulated through a captive portal (authentication), and content filtering is enforced to ensure clean browsing and prevent bandwidth abuse.

Methodology

Victory Free Wi-Fi Project is designed to ensure transparency, accountability, and continuous improvement throughout the project lifecycle. It outlines how data is collected, analyzed, and communicated across stakeholders.

1. Data Collection

- Field teams and SSFO Facilitators gather operational data from barangay sites, including uptime, usage statistics, and incident reports.
- Network performance metrics are automatically logged via the Ubiquiti Network Management System (UNMS) and UniFi Network Controller Servers.
- Maintenance teams document technical issues, repairs, and scheduled inspections.

2. Reporting Frequency

- **Daily Logs:** Captured by monitoring systems and field personnel for internal tracking.
- **Weekly Reports:** Summarize deployment progress, system performance, and minor incidents.
- **Monthly Reports:** Include detailed analytics, user feedback, maintenance summaries, and recommendations.
- **Quarterly Reviews:** CISO and other technical offices to evaluate impact and adjust strategies.

3. Report Types

- **Technical Reports:** Focus on infrastructure status, device health, and bandwidth usage.
- **Operational Reports:** Cover deployment updates, logistical movements, and team activities.
- **Incident Reports:** Document outages, troubleshooting steps, and resolution timelines.
- **User Feedback Reports:** Gather insights from barangay users to assess service quality and accessibility.

4. Communication Channels

- Reports are shared on website dashboards and group chats for effective communications.
- Critical alerts and urgent updates are escalated directly to the Project Management Office.

5. Evaluation and Follow-Up

- Reports are reviewed by the Project Maintenance Team and Project Manager/CISO Chief.
- Action plans are developed based on findings, with follow-up tasks assigned and tracked.

Figure 1. Server Management and Distribution Level Topology

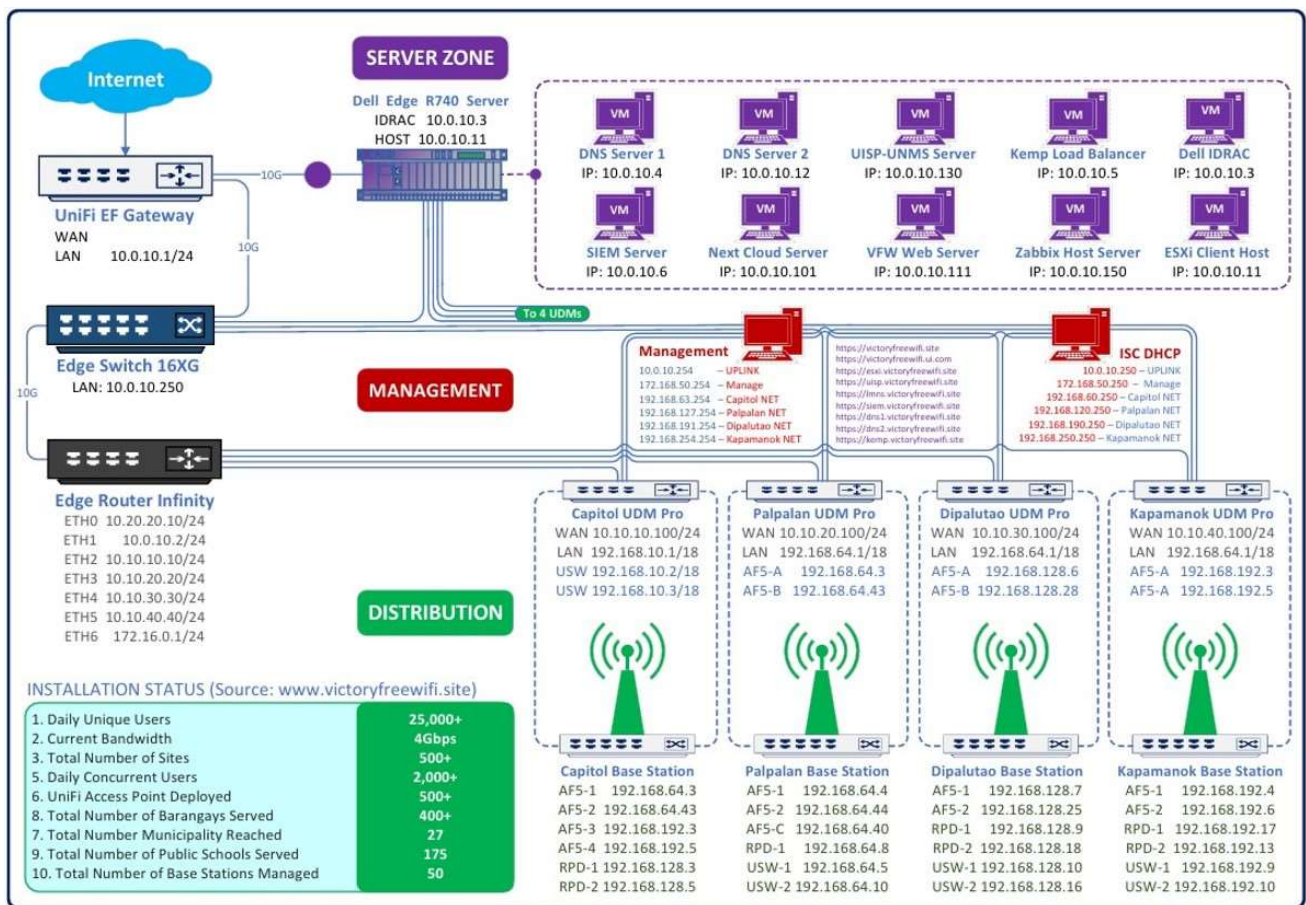
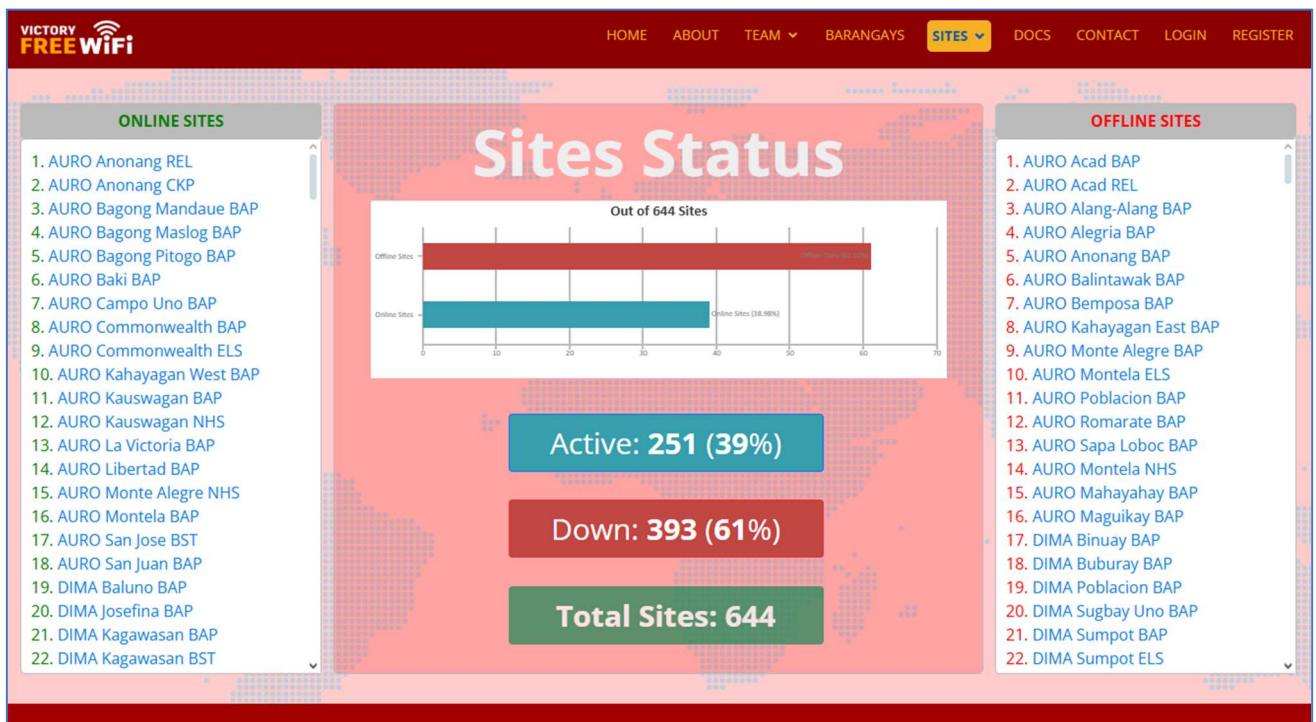
Figure 2. Sites Status: https://victoryfreewifi.net/web/sites_status_mon.php

Figure 3. Installation Report by Municipality. Over 600 sites have been established, as shown below:

#	MUNICIPALITY	BAR	WFI	CEN	SCH	OTH	TOT	GRAND TOTAL	
1.	AURORA	44	26	25	5	4	34	BAR = Total Number of Barangays	681
2.	BAYOG	28	0	0	0	0	0	WFI = Barangays with Installations	396
3.	DIMATALING	24	12	11	4	2	17	CEN = Insts on Brgys/Muns Center	393
4.	DINAS	30	25	24	16	2	42	SCH = Insts on Schools (All level)	176
5.	DUMALINAO	30	25	24	14	2	40	OTH = Insts on Other Public Places	75
6.	DUMINGAG	44	16	15	0	1	16	TOT = Total FreeWiFi Installations	644
7.	GUIPOS	17	13	13	10	3	26		
8.	JOSEFINA	14	3	3	0	2	5		
9.	KUMALARANG	18	10	9	2	3	14		
10.	LABANGAN	25	20	20	8	2	30		
11.	LAKEWOOD	14	2	1	1	2	4		
12.	LAPUYAN	26	5	6	3	0	9		
13.	MAHAYAG	29	19	16	7	0	23		
14.	MARGOSATUBIG	17	4	4	1	1	6		
15.	MIDSALIP	33	16	16	3	2	21		
16.	MOLAVE	25	21	24	4	3	31		
17.	PAGADIAN	54	53	56	26	28	111		
18.	PITOGO	15	9	10	3	2	15		
19.	RAMON MAGSAYSAY	27	22	18	14	1	33		
20.	SAN MIGUEL	18	12	12	11	0	23		
21.	SAN PABLO	28	15	14	4	2	20		
22.	SOMINOT	18	9	10	1	0	11		
23.	TABINA	15	9	10	4	8	22		
24.	TAMBULIG	31	24	24	16	1	41		
25.	TIGBAO	18	14	17	11	1	29		
26.	TUKURAN	25	10	9	7	3	19		
27.	VINCENZO SAGUN	14	2	2	0	1	3		
TOTALS		681	396	393	176	75	644		

Source: www.victoryfreewifi.net

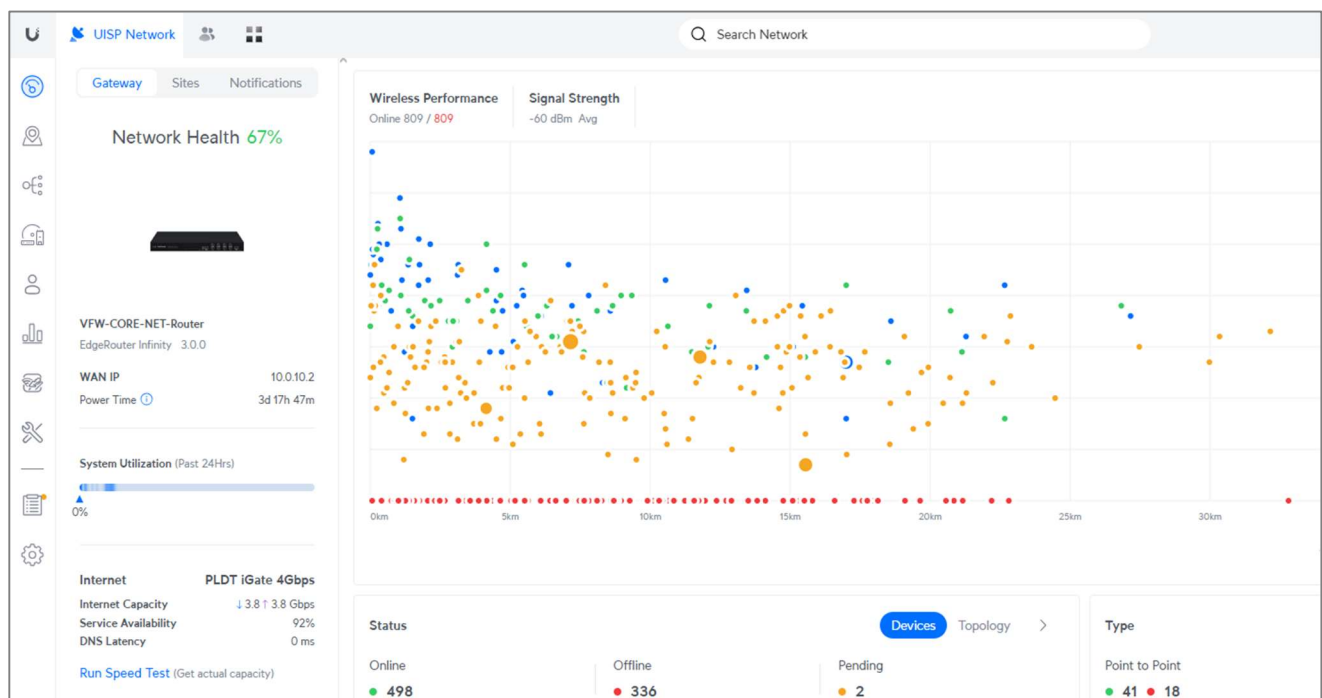
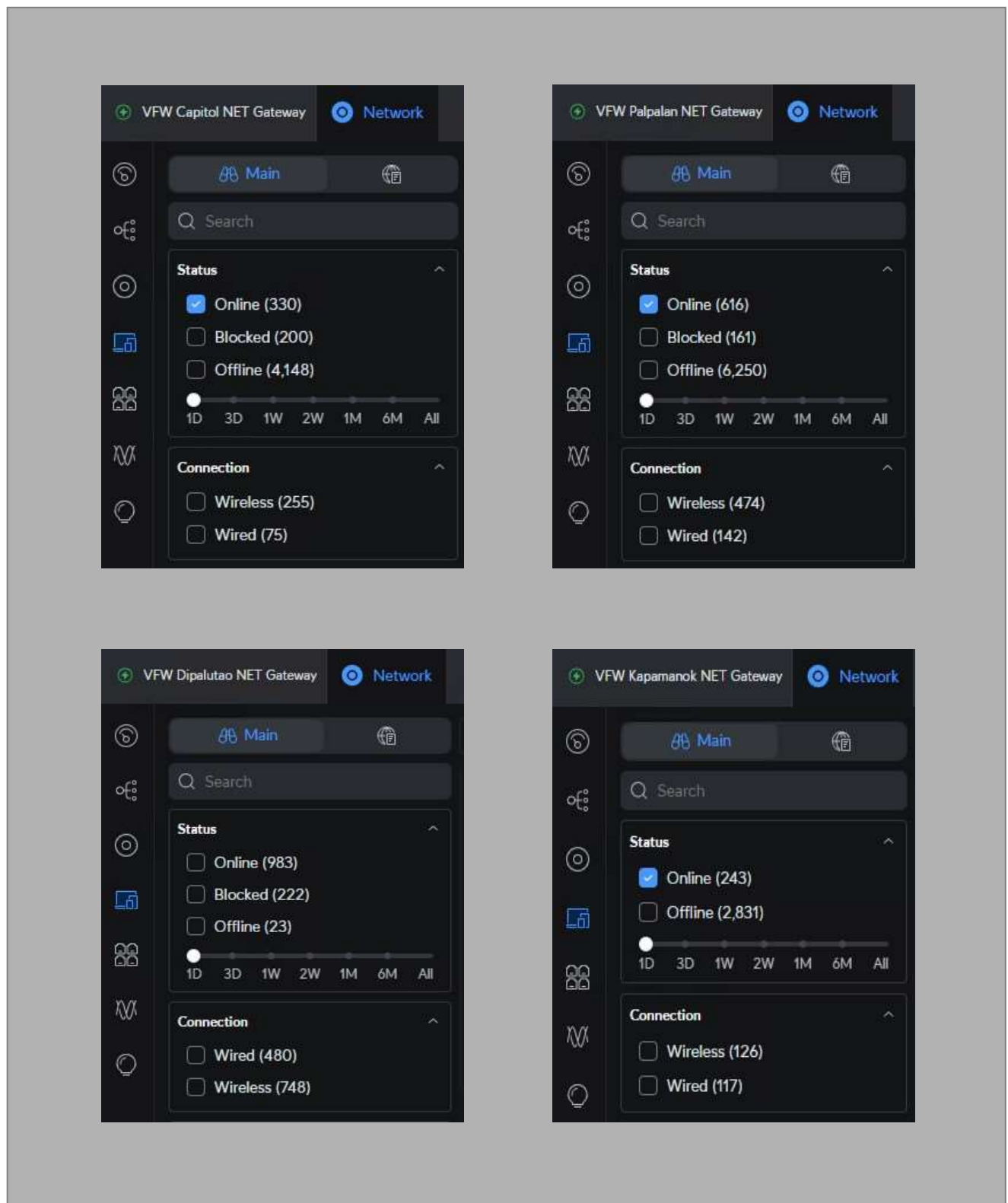
Figure 4. Network Monitoring System Dashboard <https://uisp.victoryfreewifi.net>

Figure 5. UniFi Controller Server: Displays the total number of online (concurrent) users across four network areas, indicating an average of more than 2,000 active clients.



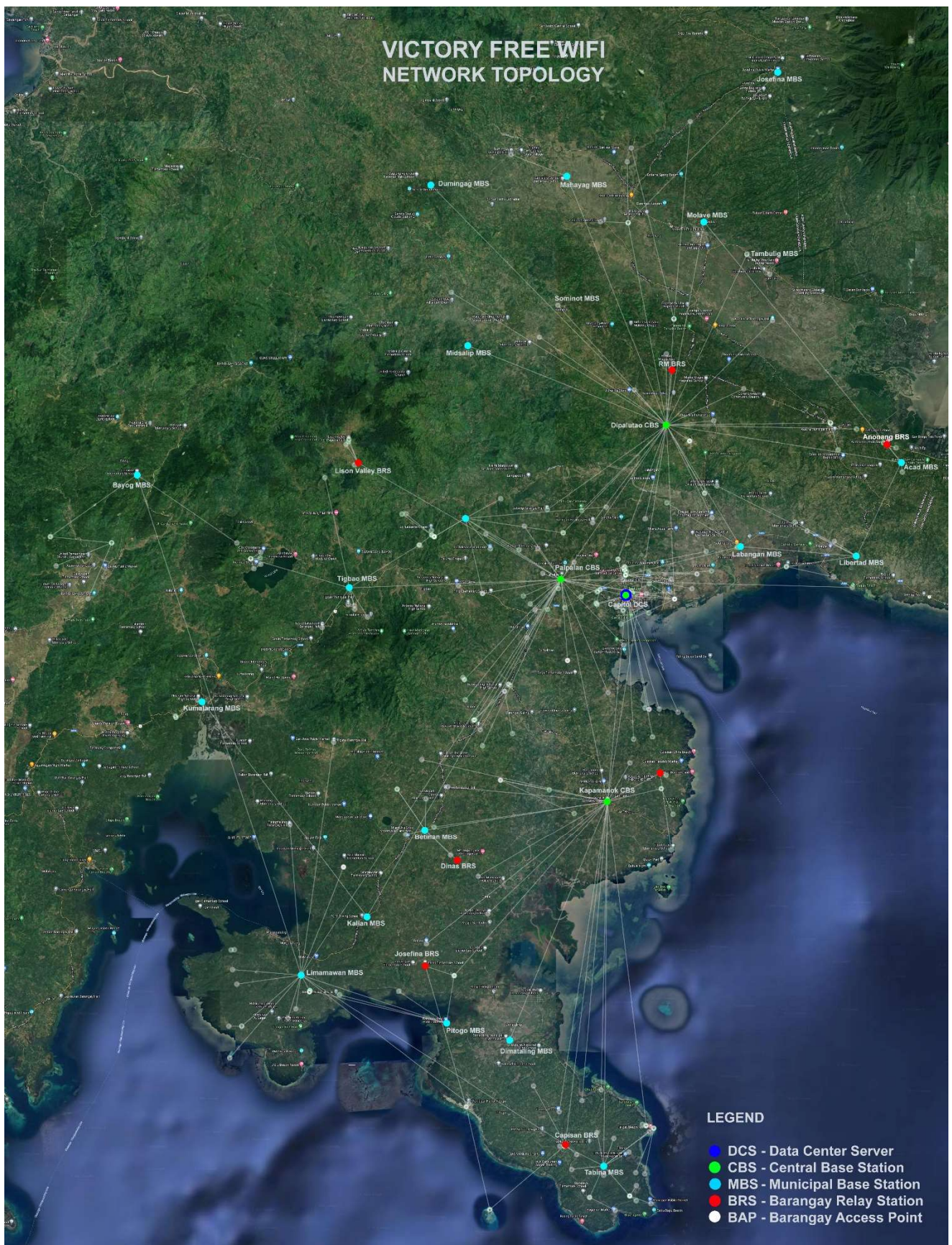
<https://victoryfreewifi.ui.com>

Figure 6. UniFi Controller Server: Displays the 24 Hour DHCP Leased across different VLANs.

Capitol NET	Name	VLAN ID	Router	Subnet	DHCP	IP Leases
	● PAGA-Group-ALL	2	VFW Capitol NET Gate...	10.100.0.0/20	Server	1707
	● PAGA-DAO-Complex	3	VFW Capitol NET Gate...	10.17.96.0/20	Server	1425
	● PAGA-Capitol-COMP	7	VFW Capitol NET Gate...	10.17.40.0/21	Server	458
	● PAGA-Bomba-Pantos	6	VFW Capitol NET Gate...	10.17.7.0/24	Server	67
	● PAGA-Nawasa-McJim	4	VFW Capitol NET Gate...	10.17.37.0/24	Server	61
	● PAGA-Balesteros-Arex	5	VFW Capitol NET Gate...	10.17.2.0/24	Server	40
Palpalan NET	Name	VLAN ID	Router	Subnet	DHCP	IP Leases
	● PAGA-Group-UAP	10	VFW Palpalan NET Gate...	10.17.0.0/20	Server	2691
	● PALP-Group-ALL	2	VFW Palpalan NET Gate...	10.100.64.0/20	Server	2024
	● DUMA-Group-UAP	4	VFW Palpalan NET Gate...	10.5.0.0/20	Server	1329
	● MARG-Group-UAP	9	VFW Palpalan NET Gate...	10.14.0.0/20	Server	597
	● GUIP-Group-UAP	5	VFW Palpalan NET Gate...	10.7.0.0/20	Server	553
	● LAPU-Group-UAP	8	VFW Palpalan NET Gate...	10.12.0.0/20	Server	388
	● TUKU-Group-UAP	12	VFW Palpalan NET Gate...	10.26.0.0/20	Server	158
	● TIGB-Group-UAP	11	VFW Palpalan NET Gate...	10.25.0.0/20	Server	59
	● KUMA-Group-UAP	6	VFW Palpalan NET Gate...	10.9.0.0/20	Server	28
Dipalutao NET	Name	VLAN ID	Router	Subnet	DHCP	IP Leases
	● DIPA-Group-ALL	2	VFW Dipalutao NET Ga...	10.100.128.0/18	Server	2008
	● TAMB-Group-UAP	14	VFW Dipalutao NET Ga...	10.24.0.0/20	Server	1285
	● MOLA-Group-UAP	10	VFW Dipalutao NET Ga...	10.16.0.0/20	Server	1147
	● RAMO-Group-UAP	12	VFW Dipalutao NET Ga...	10.19.0.0/20	Server	1124
	● AURO-Group-UAP	3	VFW Dipalutao NET Ga...	10.1.0.0/20	Server	1010
	● PAGA-Group-UAP	11	VFW Dipalutao NET Ga...	10.17.0.0/20	Server	482
	● MIDS-Group-UAP	9	VFW Dipalutao NET Ga...	10.15.0.0/20	Server	480
	● LABA-Group-UAP	7	VFW Dipalutao NET Ga...	10.10.0.0/20	Server	433
	● MAHA-Group-UAP	8	VFW Dipalutao NET Ga...	10.13.0.0/20	Server	416
	● JOSE-Group-UAP	6	VFW Dipalutao NET Ga...	10.8.0.0/20	Server	367
	● TUKU-Group-UAP	15	VFW Dipalutao NET Ga...	10.26.0.0/20	Server	361
	● SOMI-Group-UAP	13	VFW Dipalutao NET Ga...	10.22.0.0/20	Server	268
	● DUMI-Group-UAP	5	VFW Dipalutao NET Ga...	10.6.0.0/20	Server	79
	● DUMA-Group-UAP	4	VFW Dipalutao NET Ga...	10.5.0.0/20	Server	23
Kapamanok NET	Name	VLAN ID	Router	Subnet	DHCP	IP Leases
	● DIMA-Group-UAP	3	VFW Kapamanok NET ...	10.3.0.0/20	Server	832
	● DINA-Group-UAP	4	VFW Kapamanok NET ...	10.4.0.0/20	Server	749
	● SANM-Group-UAP	7	VFW Kapamanok NET ...	10.20.0.0/20	Server	609
	● PITO-Group-UAP	6	VFW Kapamanok NET ...	10.18.0.0/20	Server	527
	● KAPA-Group-ALL	2	VFW Kapamanok NET ...	10.100.192.0/18	Server	450
	● SANP-Group-UAP	8	VFW Kapamanok NET ...	10.21.0.0/20	Server	325
	● VSAG-Group-UAP	10	VFW Kapamanok NET ...	10.27.0.0/20	Server	145
	● TABI-Group-UAP	9	VFW Kapamanok NET ...	10.23.0.0/20	Server	105
	● GUIP-Group-UAP	5	VFW Kapamanok NET ...	10.7.0.0/20	Server	43

<https://victoryfreewifi.ui.com>

Figure 7. Network Topology: Illustrating how Victory Free WiFi extends connectivity to remote areas.



<https://victoryfreewifi.net>

Recommendations

Strategic

1. Expand Smart City Integration

- Begin piloting smart applications like public safety monitoring, traffic sensors, and environmental data collection using existing WiFi infrastructure.
- Partner with local universities or tech startups to co-develop low-cost IoT solutions.

2. Enhance User Experience

- Introduce multilingual captive portals to accommodate diverse local dialects.
- Implement user-friendly dashboards showing real-time network status, usage stats, and educational resources.

3. Strengthen Digital Literacy Programs

- Collaborate with DepEd and NGOs to offer regular digital skills workshops in barangays.
- Launch mobile training units to reach remote areas with hands-on tech education.

4. Boost Community Engagement

- Create a barangay-level ambassador program where trained locals promote responsible usage and report issues.
- Host community events showcasing success stories (e.g., students excelling through online learning, small businesses thriving via e-commerce).

Technical & Operational Enhancements

5. Bandwidth Optimization

- Consider dynamic bandwidth allocation based on peak usage times and priority services (e.g., education, telehealth).
- Explore content caching for frequently accessed educational materials to reduce external bandwidth load.

6. Redundancy & Resilience

- Add backup power systems (e.g., battery banks) to solar-powered stations for extended uptime during cloudy days.
- Evaluate mesh networking to improve coverage and fault tolerance in dense barangays.

7. Data-Driven Decision Making

- Use AI-based analytics to predict outages, optimize maintenance schedules, and identify high-impact expansion zones.
- Publish anonymized usage reports to demonstrate transparency and attract further investment.

Visibility & Partnerships

8. Public Relations & Storytelling

- Develop short video documentaries or infographics highlighting the transformation in specific communities.
- Share success metrics and human-interest stories on social media and local news outlets.

9. Funding & Grants

- Apply for international digital inclusion grants (e.g., ITU, World Bank, ADB).
- Offer sponsorship opportunities to private companies for branded WiFi zones or device donations.

10. Policy Advocacy

- Work with local legislators to institutionalize support for public WiFi as a basic service.
- Advocate for budget allocations in LGU annual plans to sustain and expand the network

DISCLAIMER: This document is a preliminary draft and is subject to further review and validation prior to final release.

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